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AI, hybrid intelligence, and the future of patent analytics – Key takeaways from the CEPIUG 17th anniversary conference (2025)

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ABSTRACT

The CEPIUG 17th Anniversary Conference (Istanbul, 14–16 September 2025) marked a pivotal moment in the evolution of the patent information profession. Under the theme **"Data to Wisdom: What's Next in Patent Information,"** the event highlighted how AI, hybrid intelligence, and methodological transparency are reshaping the field. Presentations explored advances in explainable AI, visual–language models, hybrid search workflows, valuation analytics, and national IP strategies, while emphasizing the continued centrality of human contextual reasoning. The discussions demonstrated that the future of IP analytics lies in integrating AI-driven scalability with human interpretative depth. This article summarizes the key insights, trends, and implications for the next generation of patent information professionals.

1. Introduction

Two years after the second edition [1], the Confederacy of European Patent Information User Groups (CEPIUG)[1]17th Anniversary Conference, held on 14-16 September 2025 at Istanbul Bilgi University, marked a significant milestone in the evolution of patent information as a professional and analytical discipline. Established in 2008, CEPIUG [2] is a non-profit European confederation [3] that brings together national patent information user groups with the mission of promoting the exchange of expertise, advancing professional standards, and supporting education and certification in patent searching and analytics across Europe. With eleven member user groups and more than 1000 professionals, CEPIUG has become a central platform for shaping best practices and professional identity within the patent information community [4].

Organized by the Turkish Patent Information User Group (TURPIUG)[2] in collaboration with Istanbul Bilgi University, the conference was developed and delivered through a comprehensive Scientific Programme Committee composed of experienced patent information professionals from academia, industry, patent offices, and international organizations. Under the theme *"Data to Wisdom: What's Next in Patent Information"* the scientific programme reflected a fundamental transformation in the profession [5] from traditional information retrieval toward knowledge synthesis, predictive analytics, and strategic foresight. Across eight thematic sessions, plenary talks, and applied case studies, the programme addressed the impact of artificial intelligence, explainable analytics, and hybrid human–machine workflows on patent searching, valuation, freedom-to-operate analysis, and innovation intelligence.

The conference was supported by major international and institutional partners, including the European Patent Office (EPO), the World Intellectual Property Organization (WIPO), the International Standards Board for Qualified Patent Information Professionals (ISBQPIP), the Patent Information Users Group (PIUG),[3] and the Patent Documentation Group (PDG).[4] In addition, a dedicated exhibition of commercial service providers brought together leading patent analytics and information vendors, enabling participants to engage directly with emerging tools and methodologies alongside the scientific discussions.

Beyond its technical scope, the conference emphasized CEPIUG's long-standing commitment to professionalization, education, and ethical practice. Particular attention was given to the Qualified Patent Information Professional (QPIP)[5] framework, continuing professional development, and the responsible use of AI in patent information work, including the presentation of the Susan Helliwell Memorial Award [6]. Bringing together more than 150 participants from 22 countries ranging from patent information specialists and R&D managers to academics, legal practitioners, and software developers the event addressed a critical need within the community: a shared, methodologically grounded forum to align technological innovation with human expertise, trust, and professional standards. In this sense, the CEPIUG 17th Anniversary Conference filled an important gap by positioning patent information not merely as a technical function, but as a strategic intelligence discipline at the intersection of data, policy, and decision-making.

[1] https://www.cepiug.org/.
[2] https://www.turpiug.org.tr/.
[3] https://www.piug.org/.
[4] https://p-d-g.org/.
[5] https://qpip.org/.

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2. Section snippets

2.1. From data to wisdom: conceptual framework

The opening sessions framed the overall transformation of patent information work from managing large data volumes to generating actionable insights. Speakers emphasized that modern IP analytics now depend on contextual interpretation rather than data abundance. Discussions revolved around how functional language extraction, semantic clustering, and data refinement can reveal technological convergence long before it appears in market behavior. The concept of *structured innovation* emerged as a unifying theme, highlighting the need to merge linguistic and functional dimensions of patents to uncover non-obvious connections between inventions.

2.2. Artificial intelligence and IP analytics

Artificial intelligence emerged as a central theme of the conference, with discussions focusing on its expanding role in patent information and IP analytics. Presentations highlight a transformative shift in intellectual property, where traditional methodologies are being superseded by Visual-Language Models (VLMs) and explainable AI (xAI). Gualtiero Fantoni *(Co-founder, Erre Quadro)* and Riccardo Apreda *(Technology Leader, Erre Quadro)* introduced a VLM-based approach that integrates image processing with Natural Language Processing (NLP). Their framework aligns textual descriptions with patent figures to facilitate technical information extraction, streamlining the synthesis of complex inventions for examiners and analysts. They also addressed the potential limitations of AI models and propose combining AI and human intelligence as a possible strategy to enable reliable, scalable, and context-aware interpretation of complex technical content. This focus on data precision was further expanded by Alexander Giesen *(Representative, Patseer)* and Gargee Patankar *(Vice President, Patseer),* who demonstrated how AI data refinement, utilizing models specifically trained on patent literature, enables context-aware discovery and automated summarization that transcends traditional keyword-based constraints. Complementing these technical advancements, Alexander Weir *(Consultant, SimIP)* showcased the strategic utility of xAI-powered business intelligence. By leveraging the Orbis and Orbis IP datasets, Weir illustrated how transparent AI valuation and competitive benchmarking can uncover "white space" opportunities and identify high-growth entities. Simon Dewulf *(CEO, Profun AI)* presented a novel framework for mining science and patents through "functional language." By treating innovation as a dynamic "Product DNA" comprised of properties and functions, Dewulf demonstrated how AI-driven innovation mapping bridges the gap between technical feasibility and strategic R&D, uncovering uncharted opportunities in the global patent landscape. Expanding the scope to a global level, Chris Harrison *(IP Analytics Manager, WIPO)* discussed how IP analytics are being transformed from raw data into actionable policy and capacity-building tools. Through initiatives like the Patent Quest game and the IP Analytics Community of Practice, Harrison emphasized the need for analysts to evolve into "data storytellers" who ensure AI remains a trusted partner rather than a "black box," particularly in developing innovation ecosystems.

The sessions further highlighted a shift in IP analytics from retrospective reporting toward predictive intelligence. Drawing on institutional and market-based case studies, speakers illustrated how patent data can serve as an early-warning mechanism for emerging technologies, policy development, and strategic investment decisions. Emphasis was placed on the growing importance of data storytelling, methodological transparency, and reproducibility in building trust in AI-assisted analytics. Collectively, these contributions underscored the emergence of a hybrid intelligence model in which AI-driven analytical capacity is combined with human contextual reasoning, positioning patent analytics as a strategic, forward-looking component of innovation intelligence.

2.3. Best Practices and Industry Applications

The “Best Practices and Industry Applications” sessions illustrated how patent information work is increasingly evolving from technical documentation toward strategic intelligence. Thorsten Zank's (*Head of Bioscience Information, BASF*) presentation framed this shift by redefining the patent professional's role from information retriever to value-adding consultant. By comparing ontologies, machine-learning classifiers, and large language models, he demonstrated that automation delivers meaningful results only when guided by human context. His examples from crop-protection and biotechnology research highlighted hybrid workflows in which expert judgment defines relevance while AI enhances analytical reach.

Linus Wretblad *(CEO, IPScreener)* addressed the historical disconnect between R&D and IP, noting that late-stage validation often leads to wasted resources. He introduced the concept of the "GenAI Patent Information Professional," a role that leverages AI to bridge this knowledge gap, allowing R&D teams to integrate IP insights early in the innovation cycle. This evolution ensures that existing technological knowledge is used to streamline research rather than reinventing known concepts. David Wanetick *(CEO, Incremental Advantage)* provided a rigorous technical perspective on asset assessment through the "Patent Valuation Gauntlet." Wanetick argued that while data, ranging from prosecution history to technology cogency, is abundant, it is insufficient without context. By applying unique metrics to examiner allowance rates and claim structures, he demonstrated how to translate complex patent statistics into definitive financial and strategic value.

Methodological transparency and data accessibility were further explored through Philippe Borne's (*Regional Representative, INPI*) comparative benchmark of Espacenet and Patentscope. His analysis showed that each platform reflects distinct design philosophies and strengths, with no single system being universally superior. Instead, best practice emerges from combining and cross-validating multiple tools within a clearly defined research strategy. This process-oriented perspective reinforced the importance of interoperability, reproducibility, and documentation in contemporary patent information practice.

Brad Buehler (*COO, Ensemble IP*) and Jonathan Skovholt *(Head of Search Operations, Ensemble IP)* extended these principles into applied search methodology through his iterative searching model. By positioning AI as a complementary “fourth pillar” alongside keyword, classification, and bibliographic searching, he demonstrated how feedback-driven workflows can improve both efficiency and relevance. His approach reframed patent searching as a continuous analytical loop, emphasizing that AI's value is realized through repeated human interpretation and refinement rather than one-time automation.

Complementing these technical perspectives, Paul Peters (*Director of Customer Success Specialists, CAS*) and Marina Flamand (*Researcher, University of Bordeaux*)situated patent intelligence within legal-scientific and policy contexts. Peters' case studies in chemical inventive-step searching underscored the continued necessity of domain expertise and contextual reasoning in assessing obviousness, even in AI-assisted environments. Flamand's empirical analysis of Chinese environmental-technology patents at the EPO highlighted how patent information supports broader policy objectives related to technological sovereignty and sustainability. Together, these contributions reinforced the view that best practice in patent information depends on hybrid intelligence, domain knowledge, and strategic interpretation across technical, legal, and geopolitical dimensions.

2.4. Human–machine collaboration

Human–machine collaboration emerged as a forward-looking theme of the conference, with discussions emphasizing artificial intelligence as a complementary component of professional judgment rather than a substitute for it. Several presentations highlighted how generative and analytical AI tools support patent information professionals by managing scale and complexity in global patent datasets, while humans retain responsibility for contextual interpretation and decision-making. Across industrial and academic examples, a shared consensus emerged that innovation in patent intelligence depends on hybrid workflows in which human expertise and machine capabilities co-evolve.

This perspective was further elaborated by İzlem Şahinkaya (*Head of IP, Sabancı University*), who examined the expanding use of AI-powered tools for clustering inventions, assessing litigation risks, and mapping technological trajectories across diverse sectors. While these systems enhance analytical reach, she emphasized that they remain context-blind with respect to legal, ethical, and strategic implications. Consequently, she argued that effective patent intelligence relies on human-in-the-loop systems, where professionals maintain interpretive control and guide AI outputs toward responsible and strategically aligned outcomes.

Empirical and workflow-oriented insights were provided by Sakari Arvela (*Co-founder, IPRally*) and Samuel Davis (*Founder, Amplified*). Arvela's Machine Patent Examiner study demonstrated that large language models can achieve substantial recall in invalidity searches, while also revealing the current limitations of automation in capturing implicit disclosures and claim nuances. Davis complemented this analysis by introducing AI agents as integrated components of IP workflows, capable of executing routine analytical tasks within transparent and auditable systems. Collectively, these contributions reinforced a central conclusion of the conference: the future of patent intelligence lies in orchestrated hybrid systems in which machines accelerate discovery, and human experts ensure meaning, accountability, and trust.

3. Education, QPIP competence, and professionalization

Professional development and qualification emerged as a recurring theme, reflecting the growing need for structured training and continuous learning in an increasingly data-intensive patent information environment. The QPIP-related discussions emphasized that digital transformation and AI adoption must be accompanied by corresponding advances in professional competence, ethics, and methodological rigor. Across sessions, mentorship, peer review, and international collaboration were highlighted as essential mechanisms for sustaining quality and trust within an increasingly automated analytical ecosystem.

This perspective was further developed by John Zabilski (*Chair of the Accreditation Committee, ISBQPIP*), who outlined the evolving structure of the QPIP program. He emphasized that the certification framework is designed not only to assess technical knowledge, but also to institutionalize a culture of competence through multidimensional examination formats combining theory, case studies, and patent landscaping. The expansion of accredited training pathways, mentoring schemes, and continuing professional development requirements reflects a broader recognition that expertise in patent information is dynamic and must be demonstrable through ongoing learning rather than accumulated experience alone.

For 2025, the ISBQPIP Committee announced the recipient of the Susan Helliwell Award, which is conferred upon the candidate achieving the highest overall performance in the 2025 QPIP examination. The symbolic statue was presented to Anne-Catherine Binot *(IP Information Specialist, Schreder)* in recognition of her outstanding examination results. The award was formally presented by Bettina De Jong *(Secretary, PDG)* at a ceremony held in Istanbul. Established in 2021, the Susan Helliwell Award honours the memory of Ms Susan Helliwell and recognises her significant contributions and enduring dedication to the establishment and development of the QPIP certification.

The impact of generative AI on education and assessment standards was addressed by Anna Maria Villa (*Head of the Landscaping Committee, ISBQPIP*), alongside organizational perspectives presented by Carolina Franciscangelis (*Senior Patent Information Specialist, Uppdragshuset*) and Fredrik Magnusson (*Senior Patent Information Specialist, Uppdragshuset*). These contributions highlighted the need to adapt examination design and training models to evaluate critical reasoning, methodological transparency, and responsible AI use. Case studies of structured internal training programs demonstrated how certification principles can be embedded within organizational learning frameworks. Collectively, these discussions reframed professionalization as an integrated ecosystem in which education, certification, and technology jointly support the integrity and future resilience of the patent information profession.

4. National perspectives: the Turkish IP landscape

Hosting the conference in Türkiye carried particular significance, highlighting the country's growing integration into the European patent information community following TURPIUG's acceptance as CEPIUG's 11th member. The session dedicated to the Turkish IP landscape demonstrated how national patent information ecosystems can align global best practices with local capacity building. Discussions addressed the rapid expansion of innovation-driven industries, the increasing institutionalization of early-stage patent searches, and the need to better harmonize academic, industrial, and legal frameworks in support of evidence-based innovation policy.

A practice-oriented perspective was provided by Onur Yolay (*Projects and IPR Manager, Boğaziçi University CT3*), who examined post-commercialization patent disputes arising from academic–industry collaborations. His case study illustrated how contractual ambiguities, ownership of improvements, and delayed payments can escalate into litigation risks if not addressed proactively. The resolution process highlighted the importance of early risk anticipation and clear contractual design, positioning sound IP governance as a critical component of sustainable commercialization and national innovation strategies.

The industrial dimension was explored by Güler Ayyıldız Dalma (*Director of IP Department, Beko Global*), who situated IP management within the broader context of geopolitical uncertainty, technological transformation, and sustainability pressures. Drawing on corporate practice, she emphasized the role of intellectual property as a strategic asset that supports resilience and long-term competitiveness. Her analysis showed how Turkish companies are increasingly adopting data-driven IP management systems, open-innovation models, and green-technology portfolios aligned with ESG objectives, thereby embedding IP strategy within broader corporate and societal goals.

Complementing these perspectives, Hatice Recber (*Co-founder, 2H Patent Law*) revisited freedom-to-operate analysis as a preventive and innovation-enabling methodology. Through applied examples, she demonstrated how systematic classification searching and iterative query refinement can identify infringement risks early in the product-development process. Her comparative assessment of public and commercial patent databases highlighted the growing technical sophistication of Turkish practitioners and their ability to combine AI-assisted tools with expert interpretation. Collectively, these contributions portrayed Türkiye's patent ecosystem as a rapidly professionalizing and globally connected environment in which legal foresight, data-driven intelligence, and collaborative practice converge to position IP as a strategic enabler of sustainable innovation.

5. Special topics (Biological/chemical/design/FTO)

The *Special Topics* session brought together a diverse set of subjects, ranging from generative AI and patent drafting to chemical, legal, and design-related aspects of intellectual property. Despite this thematic breadth, the contributions converged on a shared concern: the rapid transformation of how patent professionals create, interpret, and manage intellectual property in an increasingly data-driven environment. Andrew Samm's (*Director of Business Development, Patently*) presentation highlighted how AI-assisted patent drafting tools are reshaping patent searching by introducing greater variability in language, claim structure, and disclosure. This development, he argued, necessitates the adoption of explainable and purpose-built AI systems for search and interpretation, reinforcing the continued importance of human expertise as a safeguard against plausible but incorrect machine-generated outputs.

The organizational and collaborative implications of these changes were addressed by Benno Jensen (*Senior Manager, Clarivate*), who focused on the interface between R&D and IP functions. His analysis demonstrated how AI-enabled threat analysis and collaborative review platforms can reduce information asymmetries and improve cross-functional communication. By enabling shared visualization, annotation, and evaluation of patent landscapes, such systems support more informed freedom-to-operate decisions and help reposition IP functions as strategic partners in innovation processes rather than purely procedural controls.

Legal and regulatory dimensions were examined through contributions by John Zabilski (*Chair of the Education Committee, PIUG*) and Guido Moradei (*Chair, AIDB*). Zabilski's systematic overview of patent expiration and extension mechanisms emphasized the strategic importance of accurate legal-status information for validity assessments, licensing, and infringement analysis, particularly in regulated industries. Moradei complemented this perspective by outlining recent developments in European design law, including expanded protection for digital and virtual designs and new representation standards for three-dimensional models. Together, these contributions underscored the need for patent information systems and retrieval methodologies to evolve in parallel with legal frameworks, ensuring that analytical practices remain aligned with the growing complexity of intellectual property in the digital era.

The tools that once defined separate disciplines AI-driven drafting engines, collaborative review portals, expiration analytics, and 3D model repositories are now interdependent parts of a single, adaptive ecosystem. The session reinforced a shared conviction that in the coming decade, success in patent information work will belong to those who combine technological fluency with interpretive precision, and who can navigate a landscape where automation multiplies both capability and complexity.

6. Future of the IP information profession

The Future of the IP Information Profession session captured how the role of patent information specialists is transforming from data managers into strategic partners. Monika Bruckmann (*Head of Information Intelligence, Evonik Industries AG*) emphasized that IP experts must move “from data to dialogue,” using storytelling and contextual insight to connect technical findings with business decisions. Monique van Leijen (*QPIP, Avantium* Support *BV*) provided a practical view from a sole QPIP in a chemical company, showing that professional success increasingly relies on networking, peer exchange, and adaptability rather than team size. Victor Rudolf (*CPO, IamIP*) showcased how AI-integrated platforms, like IamIP, can return time to professionals by automating routine analysis while enhancing transparency and collaboration.

Together, the talks suggested that the profession's future lies in hybrid intelligence where analytical rigor meets communication, and technology serves human expertise. The information professional of tomorrow will not only master tools but interpret them, shaping innovation through insight and connection.

7. Cross-disciplinary collaboration and cultural exchange

One of the most distinctive strengths of this conference was its integrative atmosphere. Patent information specialists, software developers, data scientists, and legal professionals exchanged perspectives on how to improve search tools and align algorithmic design with user behavior. Beyond the technical agenda, the cultural program including the Bosphorus boat dinner and guided city tours deepened professional bonds and strengthened Türkiye's image as a bridge between East and West. Participants repeatedly noted that such cultural immersion fosters trust and collaboration as much as formal workshops do.

8. Conclusion and outlook

The CEPIUG 17th Anniversary Conference demonstrated that the future of patent information lies in *hybrid intelligence*: the seamless interplay of AI, human expertise, and global cooperation. By combining scientific inquiry, professional dialogue, and cultural experience, the event embodied CEPIUG's mission to transform data into wisdom. Hosting the event in Istanbul not only strengthened TURPIUG's international visibility but also highlighted Türkiye's rising influence within the European IP ecosystem. The next chapter of this evolution will unfold at the **CEPIUG 20**th **Anniversary Conference**, to be held in **Strasbourg in 2028**.

CRediT authorship contribution statement

Selin Özel: Writing – review & editing. **Ekrem Ayhan Çakay:** Writing – review & editing.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

References

[1] Benoit Sollie, Nigel Clarke, Bettina de Jong, Amélia Desmedt, Anna Maria Villa, Jane List, in: CEPIUG Conference (2023): a must-to-be Event for Patent Information Professionals in Europe, vol. 76, World Patent Information, 2024 102243, https://doi.org/10.1016/j.wpi.2023.102243. ISSN 0172-2190.

[2] F. Baudour, K. Gundertofte, G. Moradei, B. Stembridge, L. Wretblad, in: Update on the Confederacy of European Patent Information User Groups (CEPIUG), vol. 41, World Patent Information, 2015, pp. 41–44, https://doi.org/10.1016/j.wpi.2015.04.001. ISSN 0172-2190.

[3] A.-G. Darmont, M. Fattori, G. Gieling, A. van de Kuilen, in: Patent Information User Groups in Europe Join Together: Foundation of the Confederacy of European Patent Information User Groups (CEPIUG), vol. 31, World Patent Information, 2009, pp. 216–218, https://doi.org/10.1016/j.wpi.2008.11.005 (3)ISSN 0172-2190.

[4] S. Hantos, in: CEPIUG 10th Year Anniversary Conference, Milan, Italy, September 2018, vol. 56, World Patent Information, 2019, pp. 60–63, https://doi.org/10.1016/j.wpi.2019.01.003. ISSN 0172-2190.

[5] S. Hantos, Editorial – is patent searching a profession? World Pat. Inf. 42 (2015) 1–3, https://doi.org/10.1016/j.wpi.2015.08.004. ISSN 0172-2190.

[6] Susan Helliwell: A tribute, World Pat. Inf. 53 (2018) 45–46, https://doi.org/10.1016/j.wpi.2018.05.003. ISSN 0172-2190.

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